

Let $\bar{\mathbb{R}}$ denote the extended real numbers, $\bar{\mathbb{R}} = \mathbb{R} \cup \{\infty, -\infty\}$ where ∞ denotes *infinity* and $-\infty$ denotes *minus infinity*. We define

$$\begin{aligned} a + \infty &= \infty, \quad a - \infty = -\infty, \quad \forall a \in \mathbb{R} \\ a * \infty &= \infty, \quad a * (-\infty) = -\infty, \quad \forall a \in \mathbb{R}, a > 0. \\ 0 * \infty &= 0, \quad 0 * (-\infty) = 0. \end{aligned}$$

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$\langle u_k : k \geq 1 \rangle$ be a given sequence.

Definition: $\lim_{n \rightarrow \infty} u_n = \infty$ if for all $M \in \mathbb{R}$, $\exists n^*$ such that for all $n \geq n^*$, we have $u_n > M$.

This is equivalently expressed as $\lim_{n \rightarrow \infty} u_n = \infty$ if $\forall M \in \mathbb{R}$, $\exists n^*$ s.t. $u_n > M \forall n \geq n^*$

Definition: $\lim_{n \rightarrow \infty} u_n = -\infty$ if $\forall M \in \mathbb{R}$, $\exists n^*$ s.t. $u_n < M \forall n \geq n^*$.

Definition: $\limsup_{n \rightarrow \infty} u_n = \infty$ if for all $M \in \mathbb{R}$, $\exists n$ such that $u_n > M$.

Definition: $\liminf_{n \rightarrow \infty} u_n = -\infty$ if for all $K \in \mathbb{R}$, $\exists n$ such that $u_n < K$.

Definition: If $\lim_{n \rightarrow \infty} u_n = -\infty$, we define $\limsup_{n \rightarrow \infty} u_n = -\infty$.

Definition: If $\lim_{n \rightarrow \infty} u_n = \infty$, we define $\liminf_{n \rightarrow \infty} u_n = \infty$.

Exercises

- (a) for $n \geq 1$ let $a_n = \frac{33n+100}{27n-25}$. Show that $\lim_{n \rightarrow \infty} a_n = \frac{33}{27}$.
- (b) Consider the sequence $(x_n)_{n=1}^\infty$ defined recursively by:
 $x_1 = 1, \quad x_{n+1} = \sqrt{2 + x_n}$ for $n \geq 1$. (i) Prove by induction that $x_n < 2$ for all $n \geq 1$.
 (ii) Show that (x_n) is strictly increasing. (iii) Deduce that (x_n) converges and find its limit $\lim_{n \rightarrow \infty} x_n$.
- (c) Let $\langle a_k : k \geq 1 \rangle$ be a given sequence, for $n \geq 1$ let $s_n = \sum_{k=1}^n a_k$. Which of the following is true? Give a proof if true and counter example if not true in (1), (2), (3) below:
 (i) If $\limsup_{n \rightarrow \infty} a_n > 1$ then $\lim_{n \rightarrow \infty} s_n$ does not exist.
 (ii) If $\limsup_{n \rightarrow \infty} a_n = 0$ and $\liminf_{n \rightarrow \infty} a_n = 0$ then $\lim_{n \rightarrow \infty} s_n$ exists and is finite.
 (iii) If $\lim_{n \rightarrow \infty} s_n$ exists and is finite, then $\limsup_{n \rightarrow \infty} a_n = 0$ and $\liminf_{n \rightarrow \infty} a_n = 0$.
- (d) Show that $\lim_n a_n$ exists if and only if $\limsup_n a_n = \liminf_n a_n$.
- (e) Show that $\limsup_n (a_n + b_n) \leq (\limsup_n a_n + \limsup_n b_n)$. Give an example to show that the inequality can be strict.
- (f) Let $a_n \geq 0$ and $b_n \geq 0$ for all $n \geq 1$. Show that $\limsup_n (a_n b_n) \leq (\limsup_n a_n)(\limsup_n b_n)$. Give an example to show that the inequality can be strict.
- (g) In each of the following cases, examine if $\sum_n a_n$ is an (absolutely) convergent series. Give reasons.
 - (1) $a_n = \frac{100^n}{\sqrt{n!}}, n \geq 1$.
 - (2) $a_n = \frac{1}{n}$.
 - (3) $a_n = \frac{1}{n^2}$.
 - (4) $a_n = \frac{1}{n \log(n)}, n \geq 2$.
 - (5) $a_n = \frac{1}{n^\alpha}, \alpha > 1$.
 You may use Cauchy condensation test, after stating it precisely for (2)-(5).